MATTER-FIRST AUTOMATED DISCOVERY OF TRANSIENT SPACETIME SHORTCUTS

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We present an automated matter-first pipeline for discovering warp-precursor spacetimes: an optimizer proposes matter, a constraint solver (`GRTresna`) builds initial data, a GPU evolution code (`GRTeclyn`) advances the full 3+1 Einstein equations, and null geodesics integrated through the *time-evolving* metric test whether light arrives earlier than in flat space. MAP-Elites quality-diversity search over a 39-dimensional bicomplex Q-ball genome saturates at candidate 87 ($S = 825$); CMA-ES refinement raises the score to 852 (candidate 146) by improving confinement-gated persistence. In the headline validation run ($L = 128$, $N = 256$, three AMR levels, $t = 30$), null rays arrive 20.2% earlier than the matched flat path (5/5 rays reaching, all quality gates passed). A three-run Richardson ladder (coarse/medium/fine, $N = 192/256/384$) yields 20.4%/20.2%/20.2% with fine–medium relative difference $\ll$1%. An external ignition controller switches off at $t = 4$; because the geodesic that defines the quoted advantage launches at $t_{\rm emit} = 4$, the 20.2% timing difference is measured in freely evolving Einstein–Klein–Gordon dynamics. Repeating the evolution with the controller disabled from $t = 0$ yields evolving $f_{\rm geo} = 20.5\%$ (5/5 rays), while late-time confinement is weaker ($\sim$2% versus $\sim$11% at $t = 30$). A matched-spacing domain ladder ($L = 96$ and $L = 160$ at $h = 0.500$) agrees with the $L = 128$ baseline to within 0.12%. In all cases the channel is transient.

I. INTRODUCTION

The goal of this work is an *automatic discovery pipeline* for faster-than-light (FTL) warp precursors. We define a warp-precursor as a constraint-satisfying matter configuration that generates a transient, effective null timing advantage prior to its own dispersal: null rays between prescribed asymptotic endpoints traverse the curved spacetime in less coordinate time than the matched flat-space path. Rather than hand-crafting a metric and hoping the implied stress-energy is physical, we want a closed loop that proposes candidates, evolves them under the Einstein equations, and retains only those that pass a gauge-invariant transport test.

The standard route is *geometry-first* (equivalently metric-first): one prescribes $g_{\mu\nu}$, reads off $T_{\mu\nu} = G_{\mu\nu}/8\pi$, and studies frozen slices [3]. That recipe is poorly suited to automated search. The implied matter is rarely the output of a consistent initial-value problem, and under dynamical evolution it typically disperses or collapses, destroying the intended channel. For a quality-diversity optimizer that must evaluate hundreds of candidates, a geometry-first genome is especially hostile: most deformations of $g_{\mu\nu}$ either violate the constraints, have no local matter realization, or fail as soon as time advances—so the archive fills with numerical failures rather than competing mechanisms.

We therefore invert the loop (Fig. 1). The optimizer proposes matter; `GRTresna` (a constraint solver that returns initial data satisfying the Hamiltonian and momentum constraints) builds the initial slice; `GRTeclyn` [8, 9] (a GPU-accelerated, AMReX-based CCZ4 evolution code) advances the spacetime; and probes ask whether a null shortcut appears. In this paper "FTL" and "null shortcut" mean the same operational claim: an *effective* reduction of null travel time relative to flat space between the same asymptotic endpoints—never local superluminal motion outside the light cone. The fractional advantage is $f_{\rm geo}$ [Eq. (13)]; for example $f_{\rm geo} = 0.20$ means the curved transit is 20% shorter than the flat comparison.

That criterion is load-bearing, not cosmetic. A warp precursor is useful only if it changes what light can do between fixed distant observers. Local coordinate signal speeds, lapse/shift cosmetics, or a positive frozen-slice diagnostic can look "superluminal" while no ray actually connects the endpoints sooner once the metric evolves. Asking whether null geodesics arrive earlier than in flat space is therefore the operational definition of the search target: a gauge-invariant transport certificate that the optimizer must earn, not a post-hoc plot.

Equally important is the *scoring metric* that turns each evolution into the fitness $S(\theta)$ used by the archive [Eq. (9)]. MAP-Elites converges to whatever S rewards. In earlier campaigns with weaker objectives the search reliably *cheated*: it harvested coordinate proxies while matter dispersed ("disperse-then-warp"), banked frozen-slice positives that failed the evolving geodesic test, or climbed scores through numerical pathologies— early horizons, integrator crashes, and other simulation-failure modes that accidentally produced large proxy signals. The present objective therefore rewards only the evolving 4D shortcut and persistence, gates health terms by nontriviality, penalizes exotic excess and horizon formation, and dispersion-gates transport credit so a flying-apart cloud cannot buy a high score. Without that metric design the archive does not discover physics; it discovers loopholes.

The search engine is MAP-Elites [7], a quality-diversity algorithm that keeps one elite per cell of a behavior

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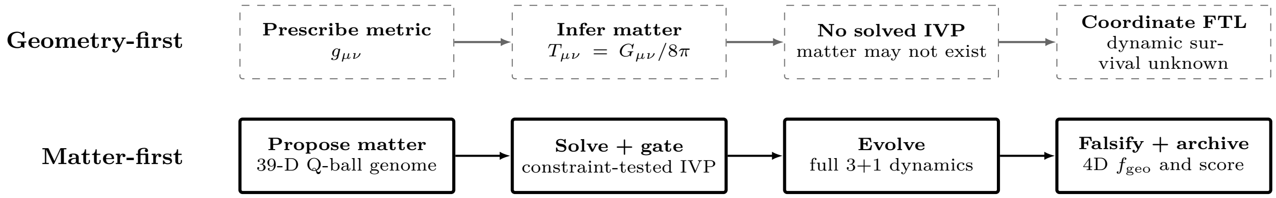

FIG. 1. Geometry-first versus matter-first discovery. Dashed: prescribe a metric and infer its stress-energy. Solid: propose matter, solve and test an initial-value problem (IVP), evolve it, and archive only candidates that pass the 4D null-transport test.

archive rather than a single global optimum. We prefer this over a pure CMA-ES hill-climb, which collapses onto one basin and discards diverse near-misses, and over reinforcement learning, which would require dense credit assignment across expensive GPU evolutions with sparse, delayed rewards. After MAP-Elites identifies a high-scoring basin, CMA-ES warm-starts from the archive champion and locally refines the genome before any high-resolution replay or resolution ladder (Sec. III). Both discovery stages run on a deliberately coarse grid chosen for evaluation throughput; production claims are deferred to the later HQ / resolution matrix (Sec. II J).

We acknowledge that the headline champion (candidate 146) relies on a phantom scalar sector and therefore does *not* bypass the standard energy-condition violations associated with FTL spacetimes. The value of this work is that an automated, matter-first Cauchy pipeline can discover, construct, and dynamically track constraint-satisfying FTL geometries—a necessary first step toward computationally searching the positive-energy sector with the same falsification stack. As a first probe of that sector, a matched canonical-only (positive-energy) search within the present ansatz finds no evolving timing advantage (Sec. VI B).

A. Methodological Rigor

Three pillars distinguish this pipeline from geometry-first studies:

1. **Initial value problem:** constraint-clean data from physical scalar sources, not coordinate scripts.
2. **Dynamic feedback:** full BSSN/CCZ4 evolution; dispersing matter collapses the channel.
3. **4D geodesic falsification:** the headline diagnostic is `ftl_geo_evolving`—null rays integrated through the *time-evolving* metric stack. Frozen-slice geodesics are retained as diagnostics but are not the transport certificate.

B. Contributions

(i) An end-to-end matter-first discovery pipeline—MAP-Elites proposal, CMA-ES local refinement of the archive champion, constraint solve, GPU evolution, and matched evolving-geodesic scoring—with an explicit reward rule applied to every evaluation (Secs. II–III).

(ii) A hierarchical campaign protocol: quality-diversity search, CMA-ES hill-climb on the top genome, then longer higher-resolution replay and a three-run Richardson resolution ladder (RC/RM/RF = coarse/medium/fine at fixed domain, varying grid spacing $h = L/N$) only on the refined champion (Sec. V).

(iii) Explicit numerical limits of the surviving candidates: dispersing exotic matter and a transient ignition-phase controller—not passenger transport (Sec. VI).

II. MATTER-FIRST PIPELINE

A. Two-code toolchain

Two open numerical-relativity codes carry the loop. `GRTresna` is a constraint solver: given a matter configuration it returns constraint-satisfying initial data $(\gamma_{ij}, K_{ij}, \phi, \Pi)$ on the initial slice, so that the Hamiltonian and momentum constraints hold before any evolution. `GRTeclyn` [8, 9] is a GPU-accelerated, AMReX-based code that evolves those data forward with the CCZ4 formulation of the Einstein equations under moving-puncture gauge. Neither code is told what metric to produce; the geometry is whatever the proposed matter sources.

B. Search Loop

One evaluation maps a parameter vector θ to a scalar fitness: $\theta \to$ initial data $\to$ `GRTeclyn` $\to$ diagnostics $\to S(\theta)$. The optimizer proposes θ, `GRTresna` solves the constraints, `GRTeclyn` evolves the spacetime on GPU, and post-processing probes reduce the evolution to the fitness $S(\theta)$ that steers the next proposal.

The matter is a canonical complex scalar field ϕ with conjugate momentum Π and potential $V(\phi)$. In the 3+1 split, a scalar field contributes an energy density ρ and a momentum density S_i that act as the sources in the Hamiltonian and momentum constraints,

$$\rho = \tfrac{1}{2}\psi^{-4}|\tilde{\nabla}\phi|^2 + \tfrac{1}{2}\Pi^2 + V(\phi), \qquad S_i = -\Pi\,\partial_i\phi, \quad (1)$$

where ψ is the conformal factor and $\tilde{\nabla}$ the conformal spatial gradient. The three terms in ρ are, in order, the gradient (spatial shape), kinetic (time variation), and potential (rest) energy of the field; S_i is nonzero only when the field oscillates in time ($\Pi \neq 0$) with a spatial gradient, which is what lets a moving scalar lump carry momentum and drive a shift. Equation (1) is therefore the precise statement of "the matter tells the geometry what to be": θ sets ϕ and Π, and `GRTresna` returns the (γ_{ij}, K_{ij}) consistent with these sources.

The word "canonical" describes the Klein–Gordon equation of motion, not the sign with which every initial lump gravitates. For each lump the optimizer stores $x_{\mathrm{ex},k} \in [0,1]$, but configuration construction rounds it to $e_k = \mathrm{round}(x_{\mathrm{ex},k})$ and sets $s_k = (-1)^{e_k}$. Thus there is no physical fractional interpolation. In the `GRTresna` constraint solve the signed independent-lump approximation is

$$\rho_{\mathrm{ID}} = \sum_k s_k\rho_k, \qquad (S_i)_{\mathrm{ID}} = -\sum_k s_k \sum_{A=1}^{2} \Pi_{A,k}\partial_i\phi_{A,k}, \quad (2)$$

with inter-lump stress-tensor cross terms omitted. The profiles themselves are then superposed into two painted complex fields—one canonical, one phantom—by the bicomplex model. Candidate 146 rounds to $s_k = (+1, -1, -1, +1, -1)$: two canonical and three phantom sources, with these signs preserved in both the constraint solve and the dynamical evolution.

C. Q-ball building blocks

The matter is built from Q-balls, and this choice was forced on us by failure. Earlier ansatz families—spherical-harmonic scalar shells and free massive (Klein–Gordon) wave packets—have no bound state and simply spread: their structural persistence collapses to a few percent well before the end of the evolution, taking any transient shortcut with them (Sec. VI A). A Q-ball is a localized, non-topological soliton of a complex scalar field: the field carries a conserved Noether charge under a global phase symmetry, and that charge holds a compact lump of field together against dispersal. This self-binding is precisely what the earlier families lacked. We choose a *complex* rather than real scalar for two reasons. First, only a complex field has the phase symmetry that supports a stable charged lump. Second, its harmonic phase rotation $\phi \propto e^{-i\omega t}$ makes $\Pi = \dot{\phi} \neq 0$, so by Eq. (1) the lump carries a momentum density S_i and can source a nontrivial shift—the ingredient a transport channel needs. Each lump starts from a spherically symmetric equilibrium profile obtained by solving the radial ODE for a

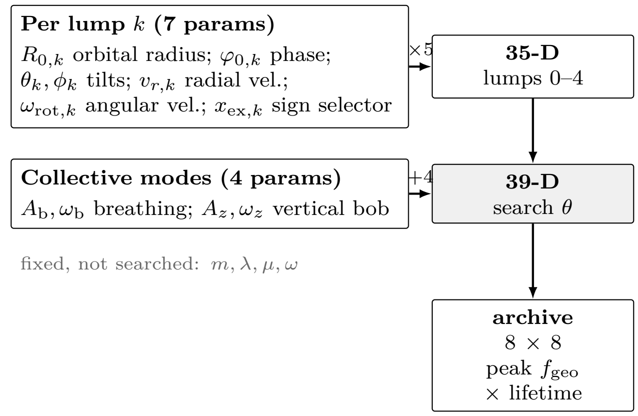

FIG. 2. Search parameterization of the trajectory Q-ball shell: $5 \times 7 = 35$ per-lump placement parameters plus 4 collective modes give the 39-D genome θ. MAP-Elites stores one elite per cell of an 8×8 archive indexed by peak shortcut and FTL lifetime. Q-ball microphysics is held fixed.

sextic potential,

$$V(|\Phi|) = \tfrac{1}{2}m^2|\Phi|^2 - \tfrac{1}{4}\lambda|\Phi|^4 + \tfrac{1}{6}\mu|\Phi|^6, \quad (3)$$

with fixed parameters $m = 1$, $\lambda = 640$, $\mu = 85333$, $\omega = 0.8$. The attractive quartic ($\lambda > 0$) provides self-binding; the repulsive sextic ($\mu > 0$) is *required* for a stable three-dimensional Q-ball—a pure quartic is the critical case and either collapses or disperses. The ratio λ^2/μ sets the minimum internal frequency $\omega_{\min} \approx 0.316$; $\omega = 0.8$ lies in the thick-wall regime where the equilibrium soliton decays to zero within $\sim$7 code units, making it compact enough to fit multiple lumps inside the evolution box. These microphysical constants are not searched.

D. Shell Parameterization

The search vector has 39 dimensions because the arrangement of the lumps, not the lump microphysics, is what is optimized. The shell contains five lumps—the maximum supported by the solver's state layout—giving enough independent tilted orbits to produce multi-stream frame dragging while keeping the search space tractable. Each lump carries seven searchable placement parameters—orbital radius R_0, phase, two tilt angles, radial velocity, angular (rotation) velocity, and a source-sign selector—giving $5 \times 7 = 35$ per-lump dimensions. Four collective variables set the shell's breathing amplitude and frequency and its vertical oscillation amplitude and frequency, for $35 + 4 = 39$ in total (Fig. 2). Although sampled continuously, the sign variable is applied as the binary selector above; it lets the optimizer choose which initial-data lumps source positive or negative stress-energy.

E. Lump orbit parameterization

Each lump moves in its own tilted orbital plane. The support center of lump k is prescribed by

$$\begin{aligned} r_k(t) &= \max[r_{\min}, R_{0,k} + v_{r,k}t + A_{\mathrm{b}}\sin(\omega_{\mathrm{b}}t)]\,, \\ \varphi_k(t) &= \varphi_{0,k} + \omega_{\mathrm{rot},k}t, \qquad z_{\mathrm{b}}(t) = A_z\sin(\omega_z t), \end{aligned} \tag{4}$$

followed by rotation into the laboratory frame through the two tilt angles (θ_k, ϕ_k). Thus each lump spirals radially (drift $v_{r,k}$), rotates in its own plane (angular velocity $\omega_{\mathrm{rot},k}$; negative values are retrograde), and shares a collective radial breathing $(A_{\mathrm{b}}, \omega_{\mathrm{b}})$ and vertical oscillation (A_z, ω_z). The range of possible orbits includes retrograde and prograde configurations, fast and slow rotation, radially drifting spirals, and nearly static scaffolds.

F. Trajectory controller (PD pump)

During evolution, a proportional–derivative (PD) controller drives the two real components of each complex field toward a moving target profile at the prescribed support center:

$$\begin{aligned} \partial_t \Pi_A|_{\mathrm{PD}} = &-wk_p(\phi_A - \phi_A^\star) \\ &- wk_d(\Pi_A - \Pi_A^\star), \qquad A = 1, 2, \end{aligned} \tag{5}$$

with gains $k_p = 12$, $k_d = 7$, and w the product of a localized envelope and a constraint-sensitive governor. This is an external, non-Lagrangian source in the scalar momentum equations; no corresponding pump stress-energy is added to $T_{\mu\nu}$, so $\nabla_\mu T^{\mu\nu} = 0$ is not exact while the pump is active. The volume-integrated pump power $P = \int \alpha\,(\Pi \cdot S_\Pi)\sqrt{\gamma}\,dV$ recorded in `collapse_diagnostics.dat` integrates to $E_{\mathrm{pump}} \simeq 1.6 \times 10^{-17}$ over the ignition window of the pump-on HQ run—consistent with floating-point noise relative to the matter amplitude ($|\phi|_{\max} \sim 7 \times 10^{-2}$). The pump therefore acts as a soft trajectory guide rather than a large energy injector. The campaign default switches the pump off at $t_{\mathrm{pump}} = 4$; scoring credits 4D geodesic transport only for $t_{\mathrm{emit}} \geq t_{\mathrm{pump}}$. A dedicated pump-free HQ twin with $t_{\mathrm{pump}} = 0$ (Sec. VI) tests whether the shortcut survives with the PD term identically off from $t = 0$.

The controller should be read as a *numerical guide* that positions the confined scaffold, not as a physical part of the reported spacetime. Two facts insulate the transport claim from its non-conservation of stress-energy. First, the pump is off throughout the scored window: the headline geodesic launches at $t_{\mathrm{emit}} = 4 = t_{\mathrm{pump}}$, so the metric it samples has been evolved by the freely conserved Einstein–Klein–Gordon system ($\nabla_\mu T^{\mu\nu} = 0$) for the entire flight. Second, the pump-free twin (Sec. VI) reproduces the advantage with the PD term identically off from $t = 0$, where $\nabla_\mu T^{\mu\nu} = 0$ holds exactly at every step. The CCZ4 constraint damping keeps the geometry and matter synchronized during the brief ignition phase, but we do not rely on that phase for any timing number.

G. Constraint gate and source-model consistency

A post-load constraint gate rejects inconsistent initial data before any GPU time is spent: the `GRTresna` Hamiltonian and momentum residuals must fall below 5%, otherwise the candidate is discarded and logged. The campaign uses the `grtresna_bicomplex_scalar` matter model, which carries two independent complex scalar fields—one canonical, one phantom—so that the per-lump gravitational signs in Eq. (2) are preserved in both the `GRTresna` constraint solve and the `GRTeclyn` evolution stress tensor $T_{\mu\nu}$. This eliminates the initial-data / evolution sign mismatch that afflicted earlier single-field campaigns. The post-load Hamiltonian and momentum gate, together with bounded CCZ4 constraint growth during evolution, therefore establishes both numerical control and source-model consistency for the bicomplex sector.

H. Phantom sector: action and evolution equations

Because the exotic sign is central to the claim, we state exactly what the `grtresna_bicomplex_scalar` model integrates. Each complex scalar is canonical, with action

$$S[\Phi] = \int d^4x\,\sqrt{-g}\,\left(-g^{\mu\nu}\partial_\mu\Phi^*\partial_\nu\Phi - V(|\Phi|)\right), \tag{6}$$

and V the sextic potential of Eq. (3). Writing $\Pi = n^\mu\partial_\mu\Phi$ for the conjugate momentum along the normal n^μ, both the canonical field Φ_+ and the exotic field Φ_- obey the *same* 3+1 Klein–Gordon system,

$$\begin{aligned} \partial_t\Phi_\pm &= \alpha\,\Pi_\pm + \beta^i\partial_i\Phi_\pm, \\ \partial_t\Pi_\pm &= \beta^i\partial_i\Pi_\pm + \alpha\left(K\,\Pi_\pm + \gamma^{ij}D_iD_j\Phi_\pm - \frac{\partial V}{\partial\Phi_\pm^*}\right) \\ &\quad + \gamma^{ij}\partial_i\alpha\,\partial_j\Phi_\pm, \end{aligned} \tag{7}$$

with D_i the spatial covariant derivative and $\alpha, \beta^i, \gamma_{ij}$ the usual lapse, shift, and spatial metric. The two sectors are therefore governed by identical, right-sign (non-ghost) kinetic operators; there is no wrong-sign kinetic term and hence none of the gradient or ghost instabilities that afflict a genuine phantom Lagrangian $\mathcal{L} = +\partial_\mu\phi\,\partial^\mu\phi^* - V$.

The exotic character enters at exactly one place: the *gravitational coupling*. The two sectors add to the Einstein source with opposite sign,

$$G_{\mu\nu} = 8\pi\left(T_{\mu\nu}[\Phi_+] - T_{\mu\nu}[\Phi_-]\right), \tag{8}$$

so Φ_- is a negative-energy (ghost-*gravitational*) source while retaining canonical, well-posed field dynamics. This sign flip is applied consistently to ρ, S_i, and S_{ij} in both the `GRTresna` constraint solve [Eq. (2)] and the `GRTeclyn` stress tensor, which is what removes the initial-data / evolution mismatch of earlier single-field campaigns. We are explicit that Eq. (8) is equivalent to reversing the effective gravitational constant for the exotic sector ($G \to -G$ for Φ_-); it is a controlled model of the negative-energy source required to violate the null energy condition, not a claim that a fundamental phantom field theory has been made well posed. The practical payoff of this choice is that the late-time dispersion reported below (Sec. VI A) is *structural*—the finite lifetime of thick-wall Q-balls together with an unbound orbital arrangement—rather than a manifestation of a phantom kinetic instability, since the fields themselves never leave the stable Klein–Gordon sector.

I. Scoring: how every evaluation is rewarded

Every evaluation in the campaign is reduced to one fitness value $S(\theta)$ by the same fixed rule, so candidates are ranked consistently. Each named quantity below is a normalized diagnostic in $[0,1]$ (penalties are negative); the score is their weighted sum,

$$S = 1000\, f_{\rm geo}^{\rm evol} + 1000\, f_{\rm geo}^{\rm op} + 600\, P_{\rm ftl} + 200\, F_{\rm op} + 200\, C_{\rm curv} + 5\, G_{\rm nt} + g_{\rm h}\, \Sigma_{\rm health} + 40\, w_x\, X_{\rm exotic} + 200\, H_{\rm hor}, \tag{9}$$

with the health block

$$\Sigma_{\rm health} = 150\,{\rm surv} + 60\,E + 40\,{\rm stab} + 30\,{\rm cmv} + 10\,\mathcal{H}_{\rm ok} + 15\,I, \tag{10}$$

gated by the nontriviality factor $g_{\rm h} \in [0,1]$ so that a trivial (flat) configuration earns no health reward. Table I lists what each term measures. Two design choices matter. First, transport dominates: the evolving 4D shortcut $f_{\rm geo}^{\rm evol}$ and its geodesic counterpart carry the largest weights, so the score chiefly tracks genuine null-transport gain. Second, the persistence and operational terms $P_{\rm ftl}, F_{\rm op}$ are *dispersion-gated*—multiplied by the fraction of matter still confined—so a candidate cannot bank a coordinate superluminal channel after its lumps have flown apart. The exotic penalty $X_{\rm exotic} < 0$ charges for energy-condition violation at campaign weight w_x. This penalty is deliberately sub-dominant to transport: at the campaign value $w_x = 0.2$ and the graded floor $X_{\rm exotic} \geq -1.6$, the exotic term reaches at most $40\, w_x |X_{\rm exotic}| \approx 13$ points, under 7% of the ≈ 200 points a 20% evolving shortcut earns ($1000\, f_{\rm geo}^{\rm evol}$). The optimizer is thus permitted to spend exotic matter freely in exchange for genuine null-transport gain, which is why the archive populates the phantom-supported sector rather than being pinned to $X_{\rm exotic} = 0$. Search-tier scores quoted later (archive and CMA-ES rankings) are values of this S on a $t = 16$ evolution: search rankings, not physical observables. The only S values reported from production ($t = 30$) runs appear in the pump-free comparison (Table VI), where they are compared solely against each other at matched duration; we never compare S across runs of different duration.

For the CMA-ES champion (candidate 146) the search-tier score is $S = 852.15$, dominated by evolving-shortcut and gated-persistence terms. The CMA-ES refinement lifted the score from the MAP-Elites plateau ($S = 825.19$, candidate 87) primarily through improved confinement-gated fitness ($0.396 \to 0.417$), not a larger raw geodesic excess (the search-tier $f_{\rm geo}$ actually decreased from 25.3% to 20.1%). This illustrates the dispersion gate at work: CMA-ES found a genome that trades a smaller instantaneous shortcut for better structural persistence.

TABLE I. Reward terms in the campaign objective [Eq. (9)], applied identically to every MAP-Elites evaluation.

symbol	rewards / penalizes	weight
$f_{\rm geo}^{\rm evol}$	4D evolving null shortcut	1000
$f_{\rm geo}^{\rm op}$	geodesic (frozen) shortcut	1000
$P_{\rm ftl}$	FTL persistence (gated)	600
$F_{\rm op}$	operational coord. FTL (gated)	200
$C_{\rm curv}$	curvature activity	200
$G_{\rm nt}$	nontrivial geometry	5
surv	structural survival	$150\, g_{\rm h}$
E	energy-condition health	$60\, g_{\rm h}$
stab	metric stability	$40\, g_{\rm h}$
cmv	co-moving stability	$30\, g_{\rm h}$
I	instability penalty	$15\, g_{\rm h}$
$\mathcal{H}_{\rm ok}$	constraint health	$10\, g_{\rm h}$
$X_{\rm exotic}$	exotic-matter penalty	$40\, w_x$
$H_{\rm hor}$	horizon-formation penalty	200

J. Search and validation stages

MAP-Elites and CMA-ES evaluate $\mathcal{O}(10^2)$ genomes, so the discovery loop uses a reduced GRTeclyn evolution configuration for wall-clock speed. Campaign env variables `GRTRESNA_EVOLUTION_L_FULL`, `GRTRESNA_EVOLUTION_N_FULL`, `GRTRESNA_EVOLUTION_MAX_LEVEL`, and `GRTRESNA_EVOLUTION_REGRID_THRESHOLD` set the search-tier AMReX inputs `max_level` and `regrid_threshold` together with the evolution box (L, N) and $t_{\rm stop}$. Archive / CMA scores are therefore *search-tier* diagnostics, not continuum claims. Only the frozen champion is re-run at production resolution for validation.

- **Stage 1 — MAP-Elites discovery (fast search grid):** quality-diversity search at $L = 64$, $N = 128$ (grid spacing $h \equiv L/N = 0.5$), `max_level`$= 1$, `regrid_threshold`$= 0.1$, and $t_{\rm stop} = 16$ (bicomplex matter; per-lump gravitational signs retained in evolution). Coarse AMR is intentional: it buys enough evaluations to fill the behavior archive. Produces a diverse archive and a provisional score leader.
- **Stage 2 — CMA-ES refinement (same search grid):** warm-start from the MAP-Elites

TABLE II. GRTeclyn evolution settings: search tier (MAP-Elites / CMA-ES) versus production validation (HQ / Richardson medium). Search uses a reduced grid for throughput; validation upgrades L, N, `max_level`, and $t_{\rm stop}$. Campaign env: `GRTRESNA_EVOLUTION_{L,N}_FULL`, `_MAX_LEVEL`, `_REGRID_THRESHOLD`.

parameter	search	HQ / RM
L	64	128
N	128	256
$h = L/N$	0.5	0.5
`max_level`	1	3
`regrid_threshold`	0.1	0.02
$t_{\rm stop}$	16	30

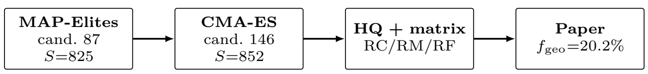

FIG. 3. Outer campaign ladder. MAP-Elites finds the diverse archive and champion (87); CMA-ES refines it to candidate 146; only the refined champion enters high-resolution replay and the resolution matrix.

champion (and optionally its nearest elites); local covariance-matrix adaptation at the *same* $(L, N,$ `max_level`$, t_{\rm stop})$ and scoring rule. Only the refined top genome is promoted further.

- **Stage 3 — production replay (HQ):** the frozen CMA-ES champion is re-solved and evolved at higher fidelity / longer $t_{\rm stop}$—typically $L = 128$, $N = 256$, `max_level`$=3$, `regrid_threshold`$=0.02$, $t_{\rm stop} = 30$—with multiple emission times and multi-ray geodesic probes. Paper headline numbers come from this stage, not from MAP-Elites.
- **Stage 4 — resolution and domain matrix:** matched coarse/medium/fine resolution runs (RC/RM/RF at fixed $L = 128$, $N = 192/256/384$) test continuum stability at the production `max_level`; matched-spacing domain runs (DS/DL at $h = 0.500$, $L = 96/160$) test boundary sensitivity on the same $t_{\rm stop}$ window.

III. SEARCH METHODOLOGY

A. MAP-Elites quality-diversity

MAP-Elites [7] is a quality-diversity search: instead of returning one champion, it maintains an 8×8 archive with at most one elite per behavior cell. A candidate displaces the occupant of its cell only if its score $S(\theta)$ [Eq. (9)] is higher. The two archive axes are peak shortcut strength and FTL lifetime, so the campaign can keep short-lived strong channels and longer weaker ones side by side. The bicomplex campaign scheduled 200 evaluations: 148 completed GPU evolution, 50 failed the post-load gate, one failed the GRTresna admission gate, and one failed the constraint solve. Nine cells were occupied (14.06% coverage). All nontrivial elites lie in the highest lifetime row, while peak strength spans the archive. The best score rose from 690.1 to 825.19 and last improved at batch 10, when candidate 87 entered cell $(7, 7)$.

TABLE III. Two-stage search hand-off: MAP-Elites champion (87) $\rightarrow$ CMA-ES refined champion (146) $\rightarrow$ HQ validation (RM). Search-tier grid: L=64, N=128, `max_level`=1, t=16. HQ grid: Table II.

stage	S	rays	peak $f_{\rm geo}$	gated fitness
MAP-Elites (87)	825.19	3/3	25.35%	0.396
CMA-ES (146)	852.15	3/3	20.12%	0.417
HQ RM (146)	—	5/5	20.21%	—

B. Archive outcome and CMA-ES refinement

The bicomplex MAP-Elites winner (candidate 87, $S = 825.19$) combines a search-tier evolving shortcut of $f_{\rm geo} = 25.3\%$ with dispersion-gated fitness 0.396, constraint-growth score 0.983, and confinement retention 0.396. CMA-ES warm-started from this genome and ran 150 evaluations (144 GPU-ok) at the same search grid, raising the score to 852.15 (candidate 146, $\Delta S \approx +27$). The improvement is not a bigger raw shortcut—the search-tier $f_{\rm geo}$ actually *decreased* from 25.3% to 20.1%—but a better confinement-gated persistence $(0.396 \rightarrow 0.417)$. CMA-ES converged to a genome with nearly static lump placement: very slow angular velocities and small breathing, producing a quasi-static scaffold that keeps matter confined longer (Fig. 8).

Search-tier confinement already falls to $\sim$40% by $t = 16$, so longer production runs are expected to stress structural persistence even if the evolving null shortcut remains positive.

IV. DIAGNOSTICS AND PROBE VALIDATION

A. The 4D evolving null-geodesic trace

The transport claim rests on tracing light rays through the simulated spacetime, not on reading coordinate speeds off a slice. A coordinate operational shortcut $F_{\rm op}$ [Eq. (12)] can flag tilted light cones, but it is partly gauge dependent and can be inflated by a coordinate choice. The authoritative probe therefore integrates the null-geodesic equations

$$\dot{x}^\mu = g^{\mu\nu} k_\nu, \qquad \dot{k}_\mu = -\tfrac{1}{2}(\partial_\mu g^{\alpha\beta}) k_\alpha k_\beta, \tag{11}$$

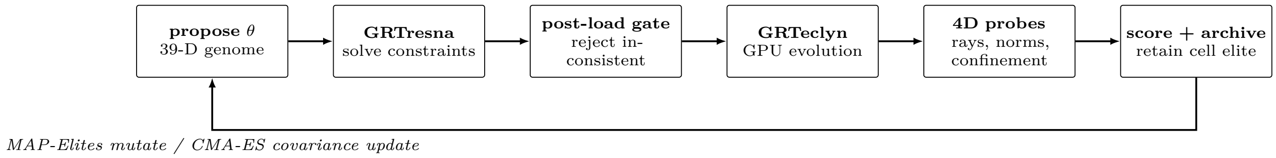

FIG. 4. Inner evaluation loop shared by MAP-Elites and CMA-ES. Failed constraint solves and post-load checks never enter GPU evolution; successful evaluations compete for archive cells (MAP-Elites) or update the CMA distribution.

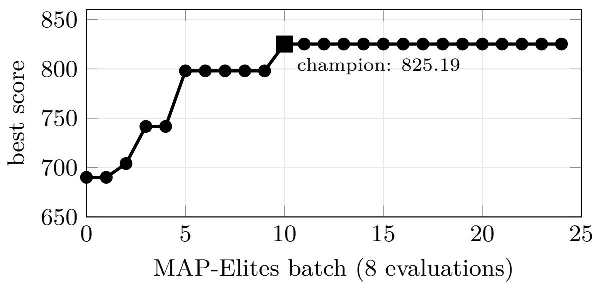

FIG. 5. MAP-Elites running best score over 200 evaluations. The champion (candidate 87) enters cell $(7, 7)$ at batch 10 with $S = 825.19$.

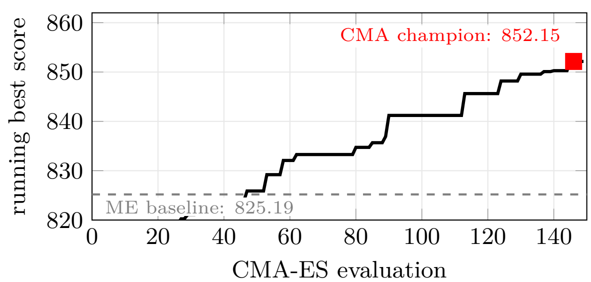

FIG. 6. CMA-ES refinement progress over 150 evaluations, warm-started from MAP-Elites champion 87. Dashed line: MAP-Elites best (825.19). The running best exceeds the baseline early and reaches 852.15 at candidate 146.

for a photon with position x^μ and momentum k_μ. "4D" means the ray is not confined to one snapshot: it is pushed through a stack of stored time slices, so at each step the metric $g_{\mu\nu}(t, \mathbf{x})$ used is the one that actually exists when the photon is at that point. This distinguishes it from a *frozen* trace, which wrongly assumes a single slice persists for the whole flight.

The construction is as follows. A ray is emitted at time $t_{\rm emit}$ from a fixed source point toward a fixed receiver. The quantities that enter the shortcut formulas are *transit durations*, not absolute grid arrival times: $t_B^{\rm curved}$ (respectively $t_{\rm min}$) is the coordinate time elapsed along the curved path after emission, and $t_B^{\rm flat}$ (respectively $t_{\rm flat}$) is the matched flat-space flight duration between the same endpoints. Absolute arrival on the grid is $t_{\rm arrival} = t_{\rm emit} + t_B^{\rm curved}$. The fractional advantage is

$$F_{\rm op} = \frac{t_B^{\rm flat} - t_B^{\rm curved}}{t_B^{\rm flat}}, \tag{12}$$

$$f_{\rm geo} = \max\left(0, \frac{t_{\rm flat} - t_{\rm min}}{t_{\rm flat}}\right), \tag{13}$$

so $f_{\rm geo} = 0$ means no shortcut and $f_{\rm geo} = 0.20$ means the curved transit is 20% shorter than the flat comparison. A launch is accepted only if its rays stay physical: the relative drift in the null Hamiltonian $g^{\mu\nu} k_\mu k_\nu = 0$ must remain below 10^{-2}, which rejects rays that have accumulated interpolation error. The simulation itself runs to $t = 30$, but that is the evolution horizon of the metric stack, not the emission schedule. A ray launched at $t_{\rm emit}$ still needs a long remaining segment of evolved geometry to reach the far endpoint (flat transit is $t_{\rm flat} = 14.4$), so late launches near $t = 30$ cannot finish a fair comparison. We therefore sweep seven early launch times, $t_{\rm emit} = 0, 2, \ldots, 12$ (five rays each), and report the largest accepted $f_{\rm geo}$—the best moment to "surf" the channel. Across the validation matrix the peak sits at $t_{\rm emit} = 4$ (just after the PD pump switches off) and has already fallen by $t_{\rm emit} = 12$, so the window covers the opening of the channel rather than cutting it off.

B. Null-constraint accuracy along the flight path

The 10^{-2} acceptance gate is a conservative safety bound, not the achieved accuracy. Two mechanisms keep the ray far tighter. First, whenever the absolute null Hamiltonian exceeds 10^{-6} the momentum is re-projected onto the local null cone before the next Runge–Kutta step, so interpolation error cannot accumulate secularly. Second, the normalized null constraint stays small between projections. Figure 7 plots the gauge-independent ratio $k_\mu k^\mu / (k^t)^2$ along the headline candidate-146 ray ($t_{\rm emit} = 4$, RM) as it crosses the matter region. It peaks at 4.1×10^{-3} near the strong-field core and has a median of 3.0×10^{-4}—roughly one to two orders of magnitude below the 10^{-2} gate—confirming that the ray remains strictly null and is not drifting into a timelike or spacelike trajectory. The complementary relative-drift diagnostic recorded by the probe, $h_{\rm rel}^{\rm max} = 2.19 \times 10^{-3}$ in RM, tells the same story.

C. Probe controls and calibration

Three controls bracket the probe's dynamic range before any candidate is evaluated.

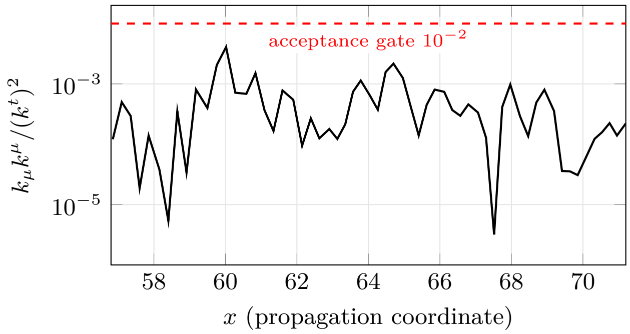

FIG. 7. Normalized null constraint $k_\mu k^\mu/(k^t)^2$ along the headline candidate-146 ray ($t_{\mathrm{emit}} = 4$, RM) from emitter A ($x = 56.8$) to receiver B ($x = 71.2$). The ray stays one to two orders of magnitude below the 10^{-2} acceptance gate (dashed), peaking at 4.1×10^{-3} in the strong-field core.

- **Minkowski (flat):** $f_{\mathrm{geo}} = 0$ by construction; verifies that the tracer introduces no spurious shortcut on trivial data.
- **Prescribed Alcubierre [4]:** a moving-bubble metric ($v_s = 0.5$, bubble radius 2, wall width 2) is built analytically and fed to the evolving tracer, yielding $f_{\mathrm{geo}} \approx 32\%$ with 5/5 rays reaching. This confirms that the 4D probe recovers a known coordinate shortcut.
- **Negative control (eval 086):** an earlier campaign candidate whose frozen-slice diagnostic shows $f_{\mathrm{geo}} \approx 5.8\%$, but whose 4D evolving HQ trace collapses to 0% (0/5 rays). This false positive motivated the design rule that the evolving trace is authoritative: frozen peaks alone do not certify transport.

The positive campaign control is the CMA-ES champion (candidate 146): every accepted launch in all three resolution-matrix runs has 5/5 rays reaching and passes the null-Hamiltonian gate.

V. RESULTS: CANDIDATE-146 WARP PRECURSOR

A. Provenance and genome

Candidate 146 is the CMA-ES-refined champion promoted from the bicomplex MAP-Elites campaign. MAP-Elites saturated at candidate 87 ($S = 825.19$); CMA-ES warm-started from that genome and after 150 evaluations raised the score to $S = 852.15$ (candidate 146, $\Delta S \approx +27$). Its provenance freezes the exact 39-dimensional genome, source checksums, solver parameters, and source commit. The configuration uses the `grtresna_bicomplex_scalar` matter model with per-lump signs preserved through both the constraint solve and the GPU evolution. The resolution-matrix RC and RM runs were metrics-only; the RF (fine) run was evolved with `GRTECLYN_FRAMES=1` and fixed color scales, from which field-map movies were produced.

B. Matter dynamics

Candidate 146 uses five lumps with source signs $s_k = (+1, -1, -1, +1, -1)$: two canonical and three phantom (exotic). The collective breathing is small ($A_{\mathrm{b}} = 0.083$, $\omega_{\mathrm{b}} = 0.429$), the vertical oscillation is moderate ($A_z = 2.32$, $\omega_z = 0.016$), and the initial radii are $R_0 = (3.69, 5.50, 2.83, 5.59, 2.61)$. All angular velocities are negative, $\omega_{\mathrm{rot}} = (-0.0017, -0.0007, -0.0031, -0.0009, -0.0049)$, so all five trajectories are retrograde but extremely slow. Radial drifts are small [$v_r = (-0.060, -0.010, 0.001, -0.005, -0.006)$]; the slow rotation and drift—roughly an order of magnitude quieter than MAP-Elites champion 87—is what CMA-ES converged to in order to maximize confinement-gated persistence.

Figure 8 shows these prescribed support-center trajectories [Eq. (4)] at four times. The Q-ball fields are initially boosted along the corresponding motion. The PD trajectory controller (Sec. II F) drives the lumps during the ignition phase ($t < 4$) and then switches off. The bounded constraints ($\|\mathcal{H}\|_2 \leq 4.73 \times 10^{-3}$, $\|\mathcal{M}\|_2 \leq 9.16 \times 10^{-4}$ in RM) show that the evolution remains numerically controlled. The support centers are the intended dynamical scaffold, not tracked density isosurfaces. The field follows them imperfectly: confinement falls from 72% to 11% and rms spread grows by a factor 3.27 by $t = 30$. Candidate 146 is therefore a transient, initially driven configuration, not a bound five-body soliton.

C. Transport mechanism

The shortcut arises from the interplay of matter momentum and geometry. Each Q-ball lump oscillates with $\Pi \neq 0$ [Eq. (1)], sourcing a momentum density S_i that enters the momentum constraint and generates a nontrivial shift β^i. The shift tilts the local light cones in the direction of the matter current; null rays propagating through this region accumulate a coordinate-time advance relative to the flat reference path (Fig. 12). All five orbits in candidate 146 are retrograde ($\omega_{\mathrm{rot}} < 0$) but extremely slow ($|\omega_{\mathrm{rot}}| \lesssim 5 \times 10^{-3}$), so the scaffold is quasi-static: CMA-ES traded a larger instantaneous shortcut (the MAP-Elites champion had $f_{\mathrm{geo}} = 25.3\%$ at search tier) for better confinement-gated persistence ($0.396 \to 0.417$), selecting slow orbits that keep matter confined longer.

The peak f_{geo} occurs at $t_{\mathrm{emit}} = 4$, precisely when the PD pump switches off and the surviving confined scaffold begins its free evolution (Sec. II F). At later emission

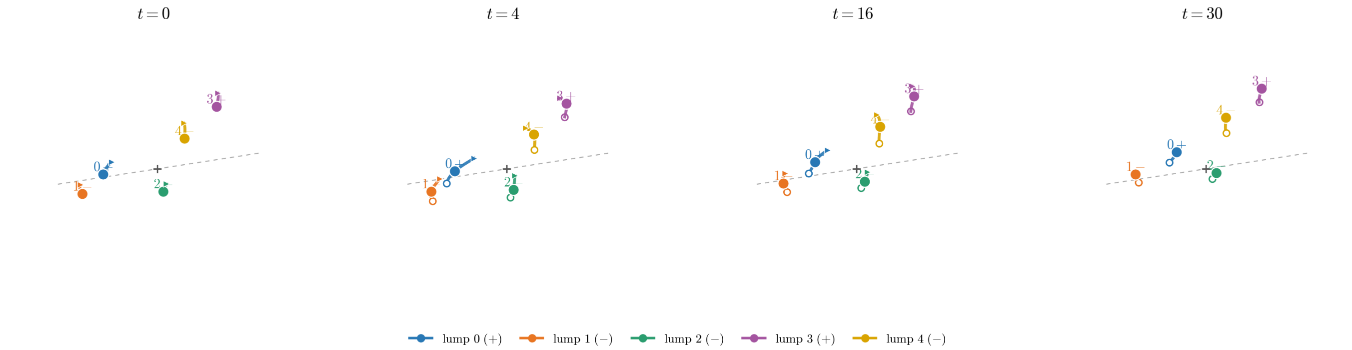

FIG. 8. Three-dimensional dynamics encoded by the frozen candidate-146 genome. Panels show the five prescribed PD-controller centers at $t = 0, 4, 16$, and 30. Filled circles are the centers at the stated times. To make the small motion legible, each local trajectory glyph covers four code-time units before and after the snapshot (where available), with its displacement about the filled center magnified by 4×; the hollow circle and arrow mark the start and end of that window. Colors and labels identify the lumps (+ canonical, − phantom), and the dashed line marks the x-directed null-probe axis. The magnification is representational: the centers are plotted at their unscaled positions. CMA-ES converged to very slow angular velocities ($|\omega_{\rm rot}| \lesssim 5 \times 10^{-3}$) and small breathing ($A_{\rm b} = 0.083$), producing a quasi-static scaffold that maximizes confinement-gated persistence.

times, the matter has dispersed further and the channel weakens—by $t_{\rm emit} = 12$ the shortcut is substantially reduced (Fig. 9). Configurations that scored lower in the archive failed for the same reason: their supporting matter dispersed and the transport channel closed with it (Sec. VI A). The dispersion gate in the scoring function [Eq. (9)] explicitly penalizes this failure mode, steering the search toward genomes that maintain confinement.

D. 4D falsification certificate

The headline claim is operational: in the RM run, null rays between the fixed endpoints arrive earlier than in flat space. The evolving-geodesic artifact reports:

peak 4D $f_{\rm geo}$	20.21%
frozen-peak $f_{\rm geo}$	29.45%
rays reached	5/5
$t_{\rm emit}$	4.00
$t_{\rm arrival}$	15.49
$t_{\rm min} = t_{\rm arrival} - t_{\rm emit}$	11.49
$t^{\rm flat}_{\rm transit}$	14.40
$h_{\rm quality\_ok}$	true ($h^{\rm max}_{\rm rel} = 2.19 \times 10^{-3}$)
search-tier peak (CMA-ES)	20.12% (3/3 rays)

Explicitly, $(t_{\rm flat} - t_{\rm min})/t_{\rm flat} = (14.40 - 11.49)/14.40 = 20.21\%$, matching Eq. (13). The peak emission time $t_{\rm emit} = 4$ coincides with the moment the PD pump switches off, suggesting that the transport channel opens as the driven ignition ceases and the matter begins its free evolution. The production HQ value agrees with the search-tier CMA-ES value to within 0.5% relative, confirming that the coarse search grid captures the essential physics. The endpoint timing corresponds to an effective $c/(1 - f_{\rm geo}) \simeq 1.25c$ relative to the matched flat path; locally the ray remains null.

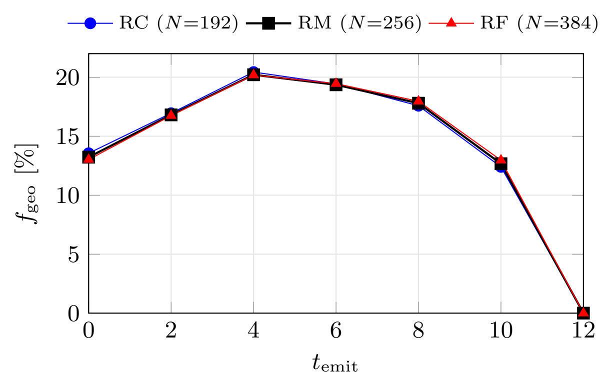

FIG. 9. Evolving null-geodesic emission sweeps ($t_{\rm emit} = 0, 2, \ldots, 12$) for all three resolution-matrix runs; the metric evolves to $t = 30$, but late launches lack enough remaining geometry for a complete transit. All runs peak at $t_{\rm emit} = 4$, just after PD pump-off, and agree within 0.26 percentage points. Every point shown has 5/5 rays reaching and passes the null-Hamiltonian quality gate.

E. Endpoint gauge and proper-time invariance

The timing advantage of Eq. (13) compares two *coordinate* transit durations between the same fixed endpoints. In moving-puncture gauge this is only a physical statement if the endpoints behave as nearly stationary, nearly flat observers; a deep lapse well ($\alpha \ll 1$) at the endpoints could in principle manufacture a coordinate advance with no proper-time content. We therefore report the metric at the two probe endpoints. The emitter and receiver sit on the propagation axis at $A = (x, y, z) = (56.8, 64, 64)$

TABLE IV. Lapse α and shift magnitude $|\beta^i|$ at the fixed probe endpoints A (emitter) and B (receiver) in RM, sampled through the emission window. The endpoints remain near-stationary ($|\beta^i| \ll 1$) and out of any deep-lapse well ($\alpha \geq 0.95$ at launch). Source: `endpoint_gauge_rm.txt`.

t	α_A	$\lvert\beta^i\rvert_A$	α_B	$\lvert\beta^i\rvert_B$
0	1.000	0	1.000	0
4	0.960	4.5×10^{-3}	0.993	3.5×10^{-2}
8	0.946	1.7×10^{-2}	1.048	9.3×10^{-2}
12	1.179	1.2×10^{-2}	1.209	1.3×10^{-2}
15	1.172	6.9×10^{-2}	1.118	7.1×10^{-2}

and $B = (71.2, 64, 64)$ in the $L = 128$ box centered on $(64, 64, 64)$ (a corridor of coordinate length $t_{\text{flat}} = 14.4$). Table IV lists the lapse α and shift magnitude $|\beta^i|$ there through the emission window. At the headline launch $t_{\text{emit}} = 4$ the endpoints are close to flat and stationary: $\alpha_A = 0.96$, $\alpha_B = 0.99$, and $|\beta^i| \lesssim 3.5 \times 10^{-2}$. The lapse never enters the dangerous $\alpha \ll 1$ regime; at late times it drifts modestly *above* unity ($\alpha \lesssim 1.2$), which if anything makes the coordinate advantage conservative, since a stationary clock with $\alpha > 1$ accumulates proper time faster than coordinate time.

To remove the residual gauge dependence entirely, we re-express the timing advantage in the proper time of a static receiver at B, whose clock ticks at $d\tau = \sqrt{-g_{00}(B)}\, dt$. Integrating the measured $g_{00}(B, t)$ from t_{emit} to each arrival, the curved signal reaches B at proper time $\tau_{\text{curved}} = 12.61$, while the matched flat-rate signal (coordinate arrival $t_{\text{emit}} + t_{\text{flat}}$) reaches it at $\tau_{\text{flat}} = 15.66$, giving a proper-time advantage

$$f^{\tau}_{\text{geo}} = \frac{\tau_{\text{flat}} - \tau_{\text{curved}}}{\tau_{\text{flat}}} = 19.5\%, \tag{14}$$

within 0.7 percentage points of the coordinate value 20.2%. The advantage is therefore not a lapse/shift artifact: it survives conversion to the physical clock of a stationary receiver.

F. Resolution and domain matrix

Validation comprises two complementary tests of candidate 146 (Table V). At fixed domain $L = 128$, a three-run resolution ladder—coarse RC ($N = 192$, $h = 0.667$), medium RM ($N = 256$, $h = 0.500$), and fine RF ($N = 384$, $h = 0.333$)—probes continuum stability. At fixed grid spacing $h = 0.500$, a domain ladder—DS ($L = 96$, $N = 192$) and DL ($L = 160$, $N = 320$)—probes sensitivity to outer-boundary placement. The medium-resolution, intermediate-domain run RM is the reference configuration quoted elsewhere in this paper. All evolutions impose non-periodic Sommerfeld (radiative) outer boundary conditions on all six faces, so the domain ladder isolates the effect of moving those radiative boundaries in or out.

TABLE V. Resolution and domain matrix for candidate 146. Every run uses three AMR levels (`max_level`$= 3$, `regrid_threshold`$= 0.02$) and terminates at $t = 30$. Peak f_{geo} is the maximum evolving 4D null-geodesic timing advantage; frozen peak is the maximum of the frozen-slice f_{geo} time series. RC/RM/RF change h at fixed L; DS/DL change L at fixed h.

run	L	N	h	peak f_{geo}	frozen peak	t_{emit}
RC	128	192	0.667	20.44%	29.15%	4
RM	128	256	0.500	20.21%	29.45%	4
RF	128	384	0.333	20.19%	29.54%	4
DS	96	192	0.500	20.19%	29.21%	4
DL	160	320	0.500	20.22%	29.44%	4

On the resolution ladder the evolving timing advantage spans 20.19–20.44%. The fine–medium relative difference is $(20.21 - 20.19)/20.21 = 0.1\%$, well below the preregistered 10% tolerance. The frozen-peak f_{geo} remains near $\sim$29.5% at all three resolutions. On the domain ladder, evolving $f_{\text{geo}} = 20.19\%$ (DS) and 20.22% (DL) differ from RM by at most 0.12%; each run returns five of five completed rays with the null-Hamiltonian quality gate satisfied. The agreement indicates that outer-boundary reflections do not set the reported timing advantage on the interval $t \leq 30$. That conclusion is consistent with a causal estimate for the $L = 128$ box: in 1+log slicing, gauge modes with characteristic speeds up to $\sim\sqrt{2}\,c$ require $t \gtrsim L/(2\sqrt{2}) \approx 45$ for a one-way trip from the outer boundary to the central region.

For unequal spacings, a Richardson estimate assumes

$$Q(h) = Q_0 + Ch^p, \qquad \frac{Q_c - Q_m}{Q_m - Q_f} = \frac{h_c^p - h_m^p}{h_m^p - h_f^p}. \tag{15}$$

The f_{geo} sequence 20.44% $\to$ 20.21% $\to$ 20.19% is monotonically decreasing and continuum-stable (fine–medium relative difference 0.1%), but the absolute gap is too small for a clean order extraction from f_{geo} alone. Applying Eq. (15) directly to the triplet with the actual spacings $(h_c, h_m, h_f) = (0.667, 0.500, 0.333)$ returns an unstable exponent ($p \approx 8.7$) whose value is set by the fine–medium difference of a single significant figure; the sequence is thus consistent with rapid convergence but does not pin down p on its own. We therefore measure the observed order on the constraint norms themselves (Fig. 10). Over the ignition window $1.5 \leq t \leq 5$, the Hamiltonian residual converges at median observed order $p_{\mathcal{H}} \approx 3.3$ and the momentum residual at $p_{\mathcal{M}} \approx 2.8$, consistent with the high-order finite-difference scheme. At late times the three resolutions already sit on top of each other to within a few percent, so the order estimator becomes noise-dominated—as expected once the continuum residual is reached. We accept the 20.2% shortcut as continuum-stable without quoting a formal extrapolated value.

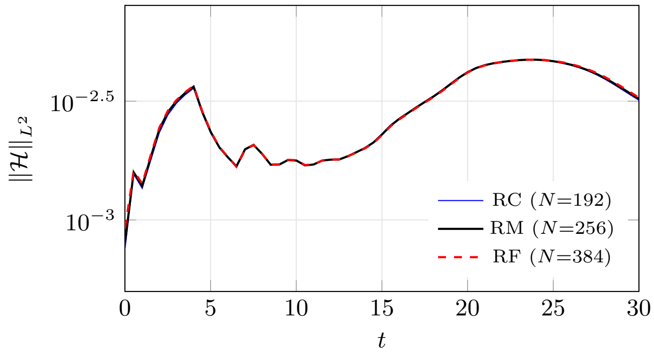

FIG. 10. Hamiltonian constraint norm across the Richardson ladder. The three resolutions overlie closely; the observed order extracted from Eq. (15) in the ignition window is $p_{\mathcal{H}} \approx 3.3$ (median).

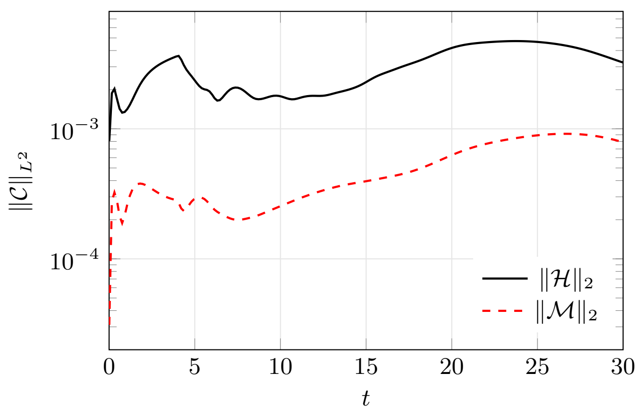

FIG. 11. Constraint norms in the headline RM run. The complete 3000-step diagnostic is plotted from a decimated table for typesetting only; maxima are evaluated from the full data.

G. Evolution constraints

For RM, the volume-weighted norms remain below $\|\mathcal{H}\|_2 = 4.73 \times 10^{-3}$ and $\|\mathcal{M}\|_2 = 9.16 \times 10^{-4}$ through $t = 30$ (Fig. 11). These are comparable to the search-tier values and remain bounded throughout the evolution. The constraint norms peak during the early driven phase ($t \lesssim 4$) and then plateau or slowly grow during the free evolution, consistent with the dispersing matter producing less active refinement at late times.

H. Frozen versus evolving shortcut, and confinement

RM's frozen-peak $f_{\rm geo}$ reaches 29.45%, compared with the evolving 4D sweep peak of 20.21%. The ∼9-percentage-point gap is physical: the frozen trace assumes that one snapshot persists for the entire ray flight, whereas the 4D trace samples the actual changing metric and therefore "sees" the channel open and then weaken during propagation. Only the latter is used for the transport claim. The frozen peak is stable across all three resolutions (29.1–29.5%), confirming that the metric snapshot quality is not resolution-sensitive.

The matter structure is not durable. In RM, confined matter fraction falls from 72% at $t = 0$ to 11% by $t = 30$, while the rms spread grows by a factor 3.27. The peak shortcut at $t_{\rm emit} = 4$ occurs just as the pump switches off, and the evolving $f_{\rm geo}$ curve falls steadily at later emission times. Successful ray transport therefore coexists with substantial dispersal: the channel is transient.

TABLE VI. Matched HQ comparison of candidate 146 with the PD controller on through $t = 4$ (RM) and disabled from $t = 0$ (PF-RM). Both runs use $L = 128$, $N = 256$, and $t = 30$. Evolving $f_{\rm geo}$ is the 4D null-geodesic timing advantage; $E_{\rm pump}$ is the integrated pump work recorded in `collapse_diagnostics.dat`; S is the final `general_ftl` score.

metric	RM ($t_{\rm pump} = 4$)	PF-RM ($t_{\rm pump} = 0$)
evolving $f_{\rm geo}$	20.21%	20.46%
rays / h quality	5/5, pass	5/5, pass
best $t_{\rm emit}$	4	12
$E_{\rm pump}$	1.6×10^{-17}	0
confined (t=0 → 30)	72% → 11%	72% → 2%
rms spread ratio	3.27	3.07
final score S	126.3	47.0

I. Evolution with the PD controller disabled

Because the PD trajectory controller (Sec. II F) is an external, non-Lagrangian source, we repeat the HQ evolution of candidate 146 with the controller disabled from $t = 0$ ($t_{\rm pump} = 0$), holding all other settings fixed ($L = 128$, $N = 256$, three AMR levels, $t = 30$). Table VI compares this pump-free integration (PF-RM) with the default pump-on twin (RM; $t_{\rm pump} = 4$).

With the PD term off, the evolving timing advantage remains $f_{\rm geo} = 20.46\%$—within 1.2% of the pump-on value—with all five rays reaching and the null-Hamiltonian quality gate satisfied. The integrated pump work vanishes identically, so the advantage cannot be ascribed to external PD energy injection; the corresponding pump-on integral is only $\sim 10^{-17}$, consistent with floating-point noise relative to the matter amplitude. The emission peak shifts from $t_{\rm emit} = 4$ to $t_{\rm emit} = 12$, indicating a slower opening of the null channel in the absence of the controller, while the peak amplitude is essentially unchanged.

The principal effect of the brief ignition-phase controller is therefore on matter retention rather than on $f_{\rm geo}$. At $t = 30$ the pump-on run still confines ∼11% of the matter, compared with ∼2% when the controller is off; the rms spread ratios remain comparable (∼3).

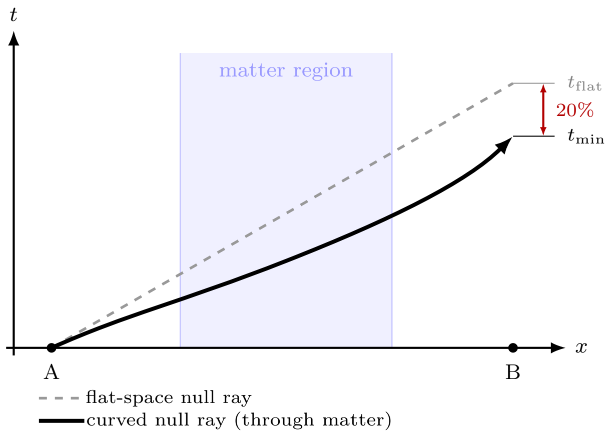

FIG. 12. Spacetime diagram of the null-geodesic shortcut. A photon emitted at A propagates toward receiver B. In flat space (dashed) the transit takes $t_{\rm flat}$. Inside the Q-ball matter region (shaded band) the geometry bends null paths so that the ray through the curved spacetime (solid) arrives at B in $t_{\rm min} < t_{\rm flat}$. The bracket on the right marks the ∼20% coordinate-time saving. The ray never leaves its local light cone.

The composite score decreases from $S = 126.3$ (RM) to $S = 47.0$ (PF-RM) for the same reason: confinement- and horizon-gated terms penalize the more rapidly dispersing pump-free evolution, while the geodesic timing advantage itself is essentially unchanged. A short PD drive in the default campaign thus delays dispersal of the supporting lumps. In both integrations the observed shortcut remains a transient precursor—a null-timing advantage coexisting with dispersing exotic matter.

J. What moves faster than light

Nothing physical moves outside its local light cone. The quoted ∼20% is a *null-geodesic timing advantage* through the evolved geometry relative to a flat reference path between the same endpoints. Coordinate light-speed maps visualize the tilted cones but do not establish the result by themselves. Figure 12 illustrates the mechanism schematically.

VI. DISCUSSION

A. The dispersion problem

Candidate 146 should be read against the many configurations that did not work. Across roughly a dozen ansatz families and several full MAP-Elites campaigns we repeatedly found the same failure mode: the supporting matter disperses and the transport channel closes with it. Spherical-harmonic shells yielded a single weak hit in ∼200 evaluations. Free massive scalar packets have no bound state and blow apart almost immediately. Bicomplex phantom lumps opened a gauge-invariant shortcut but lost ∼98% of their structure by $t = 16$ (structural persistence ≈ 0.02). Boosted and stiffened Q-ball variants improved early confinement but still sheared apart once placed on an orbit. This is why the score explicitly gates transport by confinement (Sec. III): without it, the optimizer is rewarded for a coordinate superluminal channel produced by matter that has already flown apart. CMA-ES improved confinement-gated persistence relative to the MAP-Elites champion (0.396 → 0.417), but candidate 146 still retains only about 11% of its confined matter by $t = 30$ at HQ. The search-tier confinement (∼40% at $t = 16$) falls more steeply under the longer, higher-resolution production evolution. The honest summary is that a *transient* shortcut is reproducible and robust, while a *long-lived*, well-confined channel remains an open problem.

B. Canonical-only (positive-energy) search bound

To test whether the same five-lump bicomplex Q-ball genome can produce a gauge-invariant timing advantage without phantom sources, we reran the MAP-Elites campaign of Sec. III with every per-lump exotic amplitude pinned to zero, retaining the canonical Klein–Gordon sector, retrograde-orbit bias, and postload constraint gates. Over 200 evaluations (105 GPU-accepted; 95 rejected at the postload Hamiltonian L^2 gate), the archive champion reached only $S \approx 47.4$ (eval. 130), and the evolving geodesic fraction satisfied

$$\max_{\theta\in\mathcal{A}_{\rm can}} f_{\rm geo}^{\rm evol}(\theta) = 0 \qquad (N_{\rm ok} = 105). \tag{16}$$

Several accepted genomes developed coordinate $\max v_{\rm loc} > 1$ or early apparent horizons, but none completed a null-timing advantage under the evolving-geodesic certificate. Within this ansatz and search budget, a completed $f_{\rm geo} > 0$ therefore appears to require the phantom sector used by candidate 146; the positive-energy search remains open for broader matter families and longer $t_{\rm stop}$.

C. Limitations of the matter-first approach

The matter-first strategy has complementary strengths and weaknesses relative to the geometry-first route [3]. Geometry-first prescribes $g_{\mu\nu}$ directly, so it can explore arbitrary metric deformations—including topologically nontrivial spacetimes—without specifying a matter Lagrangian. Its weakness is that the implied $T_{\mu\nu} = G_{\mu\nu}/8\pi$ is read off, not solved: there is no guarantee that the stress-energy corresponds to a consistent initial-value problem, and under dynamical evolution the configuration typically disperses or collapses. For an automated search evaluating hundreds of candidates, most deformations of $g_{\mu\nu}$ either violate the constraints or fail at the first time step, so the archive fills with numerical failures rather than competing mechanisms.

Matter-first inverts this: every candidate is constraint-satisfying by construction, and the geometry is whatever the proposed matter sources. The cost is that the search is confined to the span of the chosen matter model. Three specific limitations follow.

1. **Ansatz dependence.** The pipeline discovers only what the parameterized matter family can express. The 39-dimensional bicomplex Q-ball genome covers orbital arrangements of five solitons but cannot represent, e.g., anisotropic fluids, higher-spin fields, or engineered equation-of-state shells. Broadening the matter sector is the most direct route to new mechanism families.

2. $\mathbb{R}^3$ **spatial topology.** The `GRTresna` constraint solve and the `GRTeclyn` Cauchy evolution both operate on a topologically trivial spatial slice ($\mathbb{R}^3$ with the non-periodic Sommerfeld radiative boundaries described in Sec. V). Topological features that exist only in the full four-dimensional manifold—wormhole throats, handles, or other nontrivial $\mathbb{R}^4$ structures—cannot be represented on the initial slice and are therefore invisible to the search. Discovering such configurations would require either an extended initial-data formalism (e.g., puncture or excision methods for nontrivial topology) or a geometry-first seed that is subsequently evolved.

3. **Ignition-phase controller.** The PD pump (Sec. II F) is a non-Lagrangian external source. Evolving the same genome with $t_{\rm pump} = 0$ (Sec. VI) preserves the $\sim$20% timing advantage while reducing late-time confinement, confirming that the controller mainly retards dispersal. A fully self-consistent confinement mechanism (for example a self-gravitating boson star) would remove the need for any external drive.

The two approaches are therefore complementary: geometry-first is better suited to exploring exotic topologies and unconstrained metric deformations, while matter-first provides constraint-satisfying, dynamically evolvable candidates that can be scored by a 4D transport test. A hybrid strategy—using geometry-first motifs as seeds for a matter-first refinement loop—is a natural next step.

D. Applications and next steps

The pipeline supports (i) inverse design of null transport under energy-condition penalties, (ii) QD atlases of mechanism families, and (iii) Pareto maps of shortcut versus exotic cost and persistence. The canonical-only MAP-Elites bound of Sec. VI B already constrains the all-$+\rho$ slice of the present ansatz. Immediate priorities include longer $t_{\rm stop}$ integrations to track late-time dispersal and broader positive-energy matter families. Longer-term goals include non-dispersing matter models and confinement world-tube objectives.

TABLE VII. Static Casimir upper bounds for a 1 km silicon-photonic waveguide of width 5 μm. Amplification is the factor by which $|E_{\rm neg}|$ must increase to reach $f_{\rm geo} = 10^{-6}$. Script: `casimir_null_timing_bounds.py`.

a [nm]	$\rho_{\rm cas}$ [J/m^3]	Δt [s]	$f_{\rm geo}^{\rm bound}$	amp. to 10^{-6}
100	-4.3×10^0	6.0×10^{-62}	1.8×10^{-56}	6×10^{49}
10	-4.3×10^4	6.0×10^{-59}	1.8×10^{-53}	6×10^{46}
1	-4.3×10^8	6.0×10^{-56}	1.8×10^{-50}	6×10^{43}

E. Laboratory Casimir bounds on null-timing shortcuts

The exotic negative energy that supports candidate 146 is a continuum phantom Q-ball sector in geometric units ($G = c = 1$), not a laboratory vacuum stress. A natural question is whether a known quantum-vacuum source—the Casimir effect [10]—could produce a measurable null-timing advantage in a silicon-photonic waveguide, as suggested by cavity-warp proposals [11]. The answer, under linearized semiclassical gravity, is decisively negative.

For ideal parallel plates of separation a, the Casimir energy density is

$$\rho_{\rm cas} = -\frac{\pi^2 \hbar c}{720\, a^4}. \tag{17}$$

At $a = 10\,$nm this gives $\rho_{\rm cas} \simeq -4.3 \times 10^4\,{\rm J\,m^{-3}}$ (mass-equivalent density $\simeq -4.8 \times 10^{-13}\,{\rm kg\,m^{-3}}$). Coupling to spacetime geometry proceeds through $G/c^4 \simeq 8.3 \times 10^{-45}\,{\rm s^2\,kg^{-1}\,m^{-1}}$. For a 1 km waveguide of transverse width 5 μm and gap a, the total negative energy $E_{\rm neg} = \rho_{\rm cas}\, a\, w\, L$ induces a linearized metric perturbation of order $h \sim (G/c^4)\,|E_{\rm neg}|/L$. The corresponding timing advantage relative to a flat null path of the same coordinate length is

$$f_{\rm geo}^{\rm bound} \sim h, \qquad \Delta t \sim h\, L/c. \tag{18}$$

We emphasize that $f_{\rm geo}^{\rm bound}$ is a semiclassical upper bound on the *same timing ratio* defined in Eq. (13); it is *not* the 4D geodesic diagnostic measured on the evolving metric stack, and it is not obtained by evolving a Casimir $T_{\mu\nu}$ in `GRTeclyn`. At laboratory amplitudes the metric perturbation lies many orders of magnitude below double-precision significance on any practical numerical-relativity grid, so a full Cauchy evolution would report numerical zero, not a physical shortcut.

Table VII summarizes three gaps. Even the optimistic $a = 1\,$nm case yields $\Delta t \sim 6 \times 10^{-56}\,$s—about

37 orders of magnitude below the $\sim 10^{-19}$ s resolution of the best atomic clocks, and $f_{\rm geo}^{\rm bound} \sim 10^{-50}$ versus the $f_{\rm geo} = 20.2\%$ geodesic shortcut of candidate 146. Static Casimir cavities are therefore irrelevant for high-frequency trading, photonic interconnects, or any near-term timing application.

Closing that gap would require artificially amplifying the vacuum stress by $\gtrsim 10^{40}$ relative to Eq. (17). Proposed routes include the dynamical Casimir effect (relativistically modulated boundaries) [12], plasmonic metamaterials that concentrate vacuum fluctuations, and electromagnetically induced transparency / slow-light media that change the baseline null speed against which a geometric shortcut is measured. None of these amplifiers is implemented in the present matter-first pipeline; they remain open semiclassical modelling targets, not claims of laboratory feasibility. The practical message of this subsection is diagnostic: the automated search discovers large $f_{\rm geo}$ only because it is allowed to explore continuum exotic matter far outside known laboratory stress-energy budgets.

F. Computational cost

All runs were performed on a single 8-GPU node (NVIDIA H100 80 GB HBM3). MAP-Elites and CMA-ES each dispatch one `GRTeclyn` evolution per GPU in parallel, with concurrent `GRTresna` constraint solves running on CPU (MPI, 8 ranks per solve). A 200-evaluation MAP-Elites campaign at search resolution ($N = 128$, $t_{\rm stop} = 16$) completes in approximately two wall-clock hours; the 150-evaluation CMA-ES refinement takes a similar time. The production validation stage is sequential: the headline RM run ($N = 256$, $t_{\rm stop} = 30$, three AMR levels) fits on a single GPU; the fine Richardson run RF ($N = 384$) required three GPUs due to memory constraints. Total wall-clock time from the start of MAP-Elites to the completion of the three-run resolution matrix was approximately ten hours.

VII. CONCLUSION

We presented an automatic matter-first discovery pipeline: MAP-Elites proposes scalar configurations, CMA-ES locally refines the archive champion, the constraints are solved, the spacetime is evolved, and HQ 4D null geodesics ask whether light arrives earlier than in flat space. MAP-Elites saturated at candidate 87 ($S = 825.19$); CMA-ES improved the search score to 852.15 (candidate 146), a lift driven by confinement-gated persistence rather than a larger raw shortcut. The CMA-ES champion, a bicomplex five-lump Q-ball shell, yields $f_{\rm geo} = 20.2\%$ in the RM validation run—a 20% null timing advantage relative to flat space—and 20.2–20.4% across the resolution ladder (fine–medium relative difference $\ll 1\%$). Matched domain runs at $h = 0.500$ ($L = 96$ and $L = 160$) agree with RM to within 0.12%. The frozen-peak $f_{\rm geo}$ reaches $\sim 29.5\%$ and is stable across resolutions. All rays pass quality gates (5/5 reaching; matrix-wide $h_{\rm rel}^{\rm max} \leq 3.8 \times 10^{-3}$, with 2.19×10^{-3} in RM). The bicomplex matter model preserves per-lump canonical/phantom signs through both the constraint solve and the evolution. With the PD controller off from $t = 0$, the evolving advantage remains 20.5%, while late-time confinement is weaker; the controller mainly retards dispersal rather than generating the timing difference. When the controller is on through $t = 4$ and the geodesic launches at $t_{\rm emit} = 4$, the 20.2% advantage is measured in freely evolving Einstein–Klein–Gordon dynamics. Confinement falls to $\sim 11\%$ by $t = 30$, so the shortcut is transient. A matched canonical-only MAP-Elites campaign (all exotic amplitudes pinned to zero) found no evolving $f_{\rm geo} > 0$ in 105 accepted evaluations (Sec. VI B), so the present ansatz does not yet yield a positive-energy shortcut under the same falsification stack. Candidate 146 is therefore a reproducible, continuum- and domain-stable numerical warp precursor from an automated two-stage search, not a continuum-extrapolated or functional warp drive. Semiclassical Casimir bounds (Sec. VI E) confirm that known laboratory vacuum stresses fall $\gtrsim 40$ orders of magnitude short of the exotic budget implied by a percent-level $f_{\rm geo}$.

VIII. CODE AND DATA AVAILABILITY

The project repository contains the campaign configuration and archive, candidate-146 frozen provenance (under `runs/grtresna_promote/sources/`), and the completed validation summaries (RC/RM/RF resolution ladder, DS/DL domain ladder, and the pump-free PF-RM twin: evolving-geodesic reports, constraint histories, and confinement timeseries). The canonical-only MAP-Elites archive `qball_traj_canonical_v1` (Sec. VI B) lives under `runs/grtresna_qd/`. Figure files live in `research/neuralspacetime/figures/`; the article directory holds the reduced numerical tables used by TikZ/PGFPlots. The Casimir bound script and table (Sec. VI E) live under `research/neuralspacetime/` (`casimir_null_timing_bounds.py`, `article/data/casimir_bounds.txt`). MAP-Elites and CMA-ES trajectory files are under `runs/grtresna_qd/` and `runs/grtresna_cmaes/`.

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